Honda GX690 Methane (CNG) Conversion

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ME 444 – Combustion Engine Design

Abstract

In this study, a Honda GX690 engine platform was converted to operate on propane (C_3H_8) and compressed natural gas (CNG, CH_4) and benchmarked on a Magtrol dynamometer originally mounted to a Hatz diesel test stand. Conversion hardware includes custom carburetor adapters, high- and low-pressure fuel delivery lines, and precision-aligned shaft couplings. Propane trials employed iterative jet-adjustments at various torque setpoints to identify the optimum air–fuel ratio. Performance mapping generated complete power and torque curves and demonstrated smooth combustion and stable operation. Methane conversion efforts were impeded by insufficient supply pressure, hose length effects, and mixture control challenges. Adjustments to the fuel delivery enable smooth combustion after precise fuel adjustment. The results indicate that propane conversion of small engines is both practical and repeatable when fuel-regulator settings and mechanical alignments are tightly controlled. In contrast, CNG operation requires enhanced pressure regulation, shorter fuel lines, and robust infrastructure to overcome storage density and flow-rate limitations. This work underscores the critical interplay between gaseous fuel properties, delivery system design, and mechanical integration for alternative-fuel adaptations in portable power applications.

Introduction

Natural gas is an attractive alternative to gasoline due to its lower carbon intensity and cost per gasoline-gallon equivalent (GGE). However, small gasoline engines seldom adopt CNG conversions because of high-pressure storage and refueling infrastructure limitations. This study aims to evaluate the feasibility of retrofitting a Honda GX690, an engine commonly used in light-duty applications, to operate on CNG. With this conversion, engine power and torque will be analyzed and compared to manufacturer data, and the engine running on propane.

The technical details of our engine are shown in table 1 below.

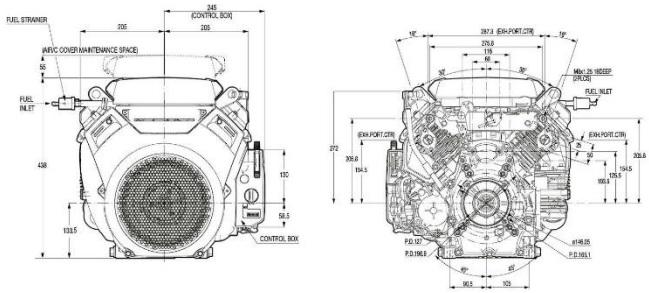

Figure 1. Schematic drawings of Honda GX690 4-stroke 2-cylinder OHV petrol engine.

Table 1. Honda GX690 Engine Parameters.

Specification	Details
Engine Type	4-stroke 2-cylinder OHV petrol engine 90° V-twin design horizontal shaft
Cylinder Sleeve Type	Cast iron sleeve
Bore x Stroke	2 / 78 x 72 mm
Displacement	688 cm³
Compression Ratio	9.3 : 1
Net Power	16.5 kW (22.1 HP) / 3600 rpm
Cont. Rated Power	11.5 kW (15.4 HP) / 3000 rpm 13.0 kW (17.4 HP) / 3600 rpm
Max. Net Torque	48.3 Nm (4.93 kgfm) / 2500 rpm
Ignition System	Digital CDI with variable ignition timing
Carburetor	2-barrel, fuel cut solenoid
Lubrication	Full-pressure
Starter	Electric
Fuel Cons. at Cont. Rated Power	6.7 L/h - 3600 rpm
Engine Oil Capacity	1.7 Liter
Dimensions (L x W x H)	429 x 410 x 438 mm
Dry Weight	44.6 kg

Background and Motivation

The increasing regulatory and societal pressures to reduce greenhouse-gas and criteria-pollutant emissions have driven significant interest in adapting small, spark-ignited engines to operate on cleaner, gaseous fuels. Propane (C_3H_8) and compressed natural gas (CNG, CH_4) both offer substantial emissions benefits over conventional gasoline including lower particulate matter, reduced carbon monoxide, and lower unburned hydrocarbons, all while leveraging existing carbureted engine architectures with minimal mechanical modification. Portable power applications, including emergency generators, agricultural equipment, and off-grid machinery, represent a large segment of the small-engine market; converting these engines to run on propane or CNG can yield immediate environmental benefits without the high capital investment of large stationary systems.

The fundamental driver of engine performance is the amount of chemical energy released during combustion and how effectively that energy is converted into mechanical work. Each fuel's intrinsic energy per unit mass (its heating value) and the density with which it can be inducted into the cylinder determine the mass of fuel–air mixture available for ignition each cycle. Propane and methane are gases at ambient conditions and therefore must occupy a larger volume in the intake manifold to deliver the same chemical energy as liquid gasoline. In a naturally aspirated engine, this means less total energy can be drawn into the cylinder on each intake stroke compared to gasoline, which can limit peak torque and power unless compensated for by enhanced induction or ignition timing adjustments.

In this particular GX690 application, we expect propane to deliver nearly the same peak power as gasoline, marginally reduced by its gaseous volumetric requirements. However, we expect methane's lower volumetric energy density to result in a noticeably reduced peak output, possibly requiring intake or timing modifications to recover performance.

To compare such dissimilar fuels on a fair basis, the gasoline-gallon equivalent (GGE) framework expresses fuel consumption and output in terms of the energy contained in one gallon of gasoline. By normalizing fuel use to a common energy standard, GGE abstracts away differences in physical state and storage pressure, enabling direct assessment of how efficiently an engine converts the chemical energy of propane or methane into useful work relative to gasoline. This is a key metric in understanding how much fuel volume is needed, which becomes a major constraint when designing engines to run off gaseous fuels.

Table 2. Relevant properties of Methane and Propane, tabulated.

Property	Methane (CH_4)	Propane (C_3H_8)
Stoichiometric Air:Fuel Mass Ratio (AFR)	17.2 : 1	15.5 : 1
Lower Heating Value (LHV)	≈ 50 MJ/kg	≈ 46 MJ/kg
Gas Density (25 °C, 1 atm)	0.656 kg/m³	1.88 kg/m³ (approx.)
Boiling Point	−162 °C	−42 °C
Storage State & Pressure	Compressed (CNG) at up to 3600 psi	Liquefied (LPG) at around 120–150 psi
Gasoline Gallon Equivalent (GGE)	0.26 (at 3600 PSI)	.74

To analyze the expected output from each of our fuels at stoichiometric conditions, we can use our engine parameters and the physical and thermochemical properties of our fuels to estimate the amount of energy released during a combustion stroke in our engine for each fuel.

Assuming a perfect fuel conversion efficiency, the net work done in a combustion cycle is equal to

$$W_{net} = m_f Q_{LHV}$$

, or simply put the mass of fuel in the combustion chamber times the lower heating value of the fuel.

The mass of fuel in the combustion chamber at stoichiometric conditions (assuming perfect volumetric efficiency) is determined by the cylinder volume, the stoichiometric air-fuel ratio, and the densities of air and fuel. This can be seen below.

$$V_{cyl} = V_{air} + V_{fuel}$$

$$V_{cyl} = \frac{m_{air}}{\rho_{air}} + \frac{m_{fuel}}{\rho_{fuel}}$$

$$V_{cyl} = \frac{(AFR)_s m_{fuel}}{\rho_{air}} + \frac{m_{fuel}}{\rho_{fuel}}$$

Substituting in our own cylinder volume, tabulated values for stoichiometric AFRs, and densities based on standard temperature and pressure, the mass of fuel was solved for and the following values of net work per combustion cycle (per cylinder) were obtained.

Table 3. Calculated work outputs for different fuels

Fuel	Net work per cycle per cylinder (kJ)
Methane	1.07
Propane	1.16

Gasoline*	1.19

**Value for gasoline estimated assuming gas injected as liquid (air volume is approximately equal to cylinder volume, and heat of vaporization subtracted from work output) and with AFR = 14.7, Q_{LHV} = 43 MJ/kg.*

The limiting factor in every case is the air volume in the cylinder. Methane and propane enter the cylinder as a gas, and their own volume takes away from the volume of air that can be in the cylinder. Additionally, the higher stoichiometric air to fuel ratio of propane and methane means that even less fuel can be burned per cycle. This is why methane combustion outputs the least work per cycle despite having the largest lower heating value.

Thermodynamic efficiency affects these numbers equally, and so even in our real-world testing we expected gasoline to produce the most power, followed by propane, followed by methane as long as we kept our mixture stoichiometric.

In practice, most engines are not run with true stoichiometric mixtures, but at operating points with slight excess air (lean condition) or slight excess fuel (rich condition). The equivalence ratio is a parameter which captures how far from stoichiometric the current operating point is. It is defined as:

$$\lambda = \frac{AFR}{AFR_{stoich}}$$

Engine Test Stand

In order to collect quantitative performance data of the Honda GX690, it was necessary to construct a mount for the engine so that it could be coupled to the Magtrol WB-series 15kW dynamometer, which was previously being used to measure the output from a Hatz D-Series single-cylinder diesel engine.

After running the Hatz diesel engine to verify dynamometer functionality, we decoupled it from the dynamometer and then removed it from the test stand table. Successfully removing the coupling proved difficult, as it took complete deconstruction to detach from the dynamometer.

Once the Hatz engine was removed, we found a compatible coupling to connect the GX690 engine onto the dynamometer. From there, we created a temporary wooden engine mount to match the height of the dynamometer. We began by aligning the shafts in the horizontal direction, then sanding down the mount until the shafts were perfectly aligned.

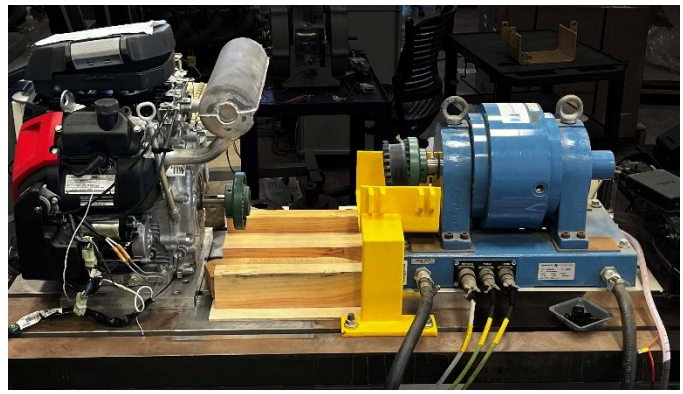

Figure 2. Image of GX690 test stand before attaching the engine to the mount.

Propane System Components

As propane is stored as a liquid, fuel delivery requires low-pressure components. Our propane conversion system consisted of a propane tank fitted to a supplemental propane regulator. This smaller regulator is attached to a high-pressure line that leads into a low-pressure "pancake" regulator. [Note that this line does not require high pressure, as this was reused for our methane setup.] The pancake regulator feeds into a carburetor attachment via a temperature resistant line. Fuel is mixed with air radially through an attachment to the engine's carburetor intake.

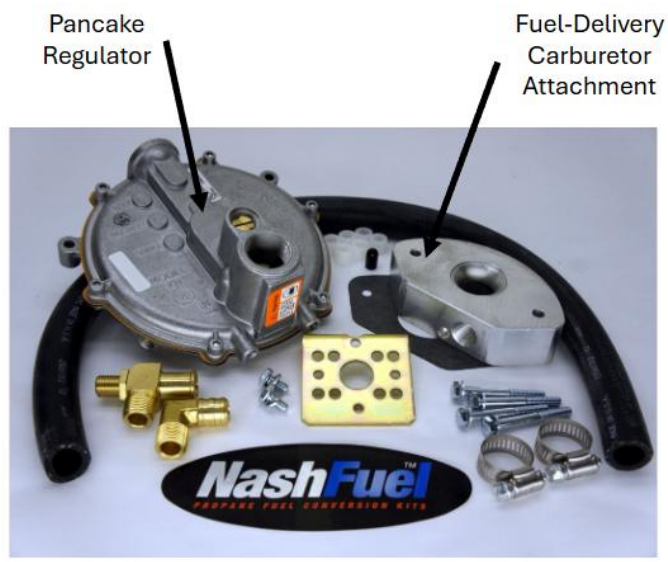

Figure 3. Diagram of propane conversion components, courtesy of NashFuel.

Methane (CNG) System Components

The CNG converted setup is quite similar to the propane one. From the tank of methane, fuel at >1000psi is run through a high-pressure regulator to limit our "injection" pressure. This regulator connects to a high-pressure line and back to the pancake regulator used in the propane setup. Again, a temperature-resistant line connects the pancake regulator to the carburetor attachment shown in Figure 3.

Initially, we were unable to achieve high enough fuel pressure to achieve continuous combustions. The tank of methane was originally connected to a gas line in the lab that ran down the wall. We tried to use this system and attached our hose to one of the exit valves.

After many failed attempts using this setup, we bypassed the lab-integrated system and directly connected our fuel line to the exit of the high-pressure regulator. With careful fuel adjustment, we were able to sustain combustion using this system.

Future System Upgrades

In the next iteration of this setup, we wish to better monitor and regulate the pressure of fuel entering the engine. We would like to replace the pancake regulator with a more robust one with pressure gauges and a more appropriate valve to adjust the restriction to flow.

Additionally, we were hoping to attach a F5000 Vehicle Exhaust Gas Analyzer to our setup to gain emissions data and view our air-fuel ratio (AFR). Unfortunately, at the time of testing, this was not functioning. Moving forward, we would hope to get results from this, to compare the environmental impacts of both fuels and better tune the AFR.

Results

Propane Fuel Adjustments

As the exhaust gas analyzer was non-functional, we were forced to find an alternative method to set the ideal AFR. With the dynamometer attached, we ran the engine at two constant torques, 15 and 20 ft-lbs.

Figure 4. Fuel adjustment screw after pancake regulator.

At each torque, we ran the engine, then adjusted the fuel adjustment screw to restrict the flow of fuel. Once the engine began to die due to the lean mixture, we recorded the number of threads exposed on the screw and set that as the lower limit. We then measured the engine speed and power. After taking data, the screw was backed-out by a quarter turn, and speed and power were recorded once again. This process continued until the engine began to die again and noted this point as the upper limit. The results from these tests were plotted (shown in figures 5 and 6) and the screw position at peak power was recorded. We determined the average of these positions was the ideal fuel flowrate for the engine, at +¾ turn from the lower fuel limit. This equated to around 7.5 exposed threads on the fuel adjustment screw.

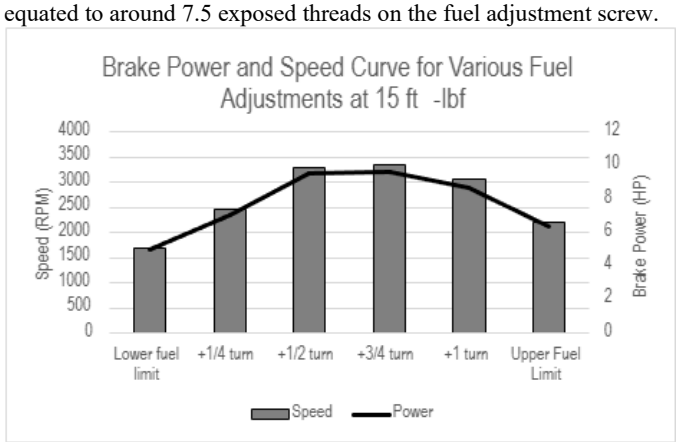

Figure 5. Engine speed and power for various fuel screw positions. All measured at a torque of 15 ft-lbs.

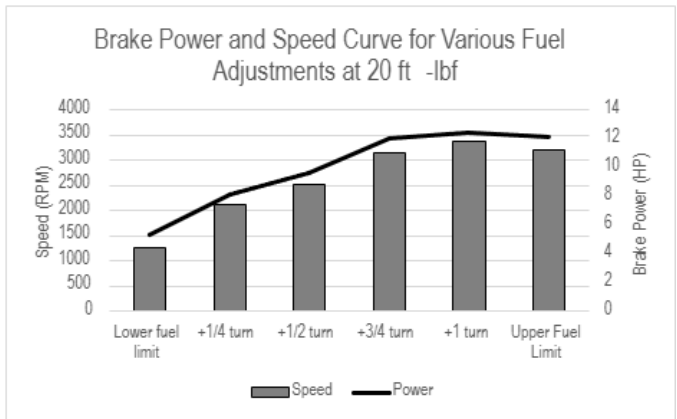

Figure 6. Engine speed and power for various fuel screw positions. All measured at a torque of 20 ft-lbs.

Propane Power Curve

Once the fuel adjustment screw was tuned to maximize engine performance, a power test was conducted across a torque sweep.

With propane, we found a peak power output to be around 9.5 BHP at 2300 RPM. This is significantly less than the gas as the engine is rated for a BHP of 22.1 at 3600 RPM. This data is portrayed graphically in Figure 7. We believe that the fuel is still too lean. Because of the components we used we did not have much control over the fuel flow and that would explain our lower performance values.

Upon testing, we discovered that the Magtrol dynamometer we were using to quantify the performance of the engine is slightly underpowered for our use case. This arose when the dynamometer could no longer increase torque, yet the engine did not appear to be about to stall.

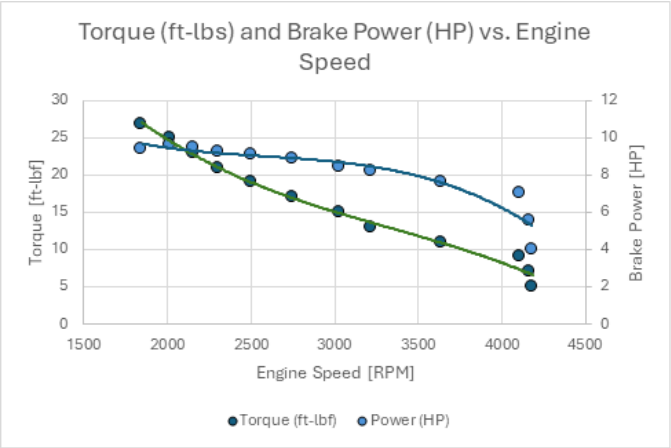

Figure 7. Engine torque and power vs speed for GX690 engine running with propane fuel.

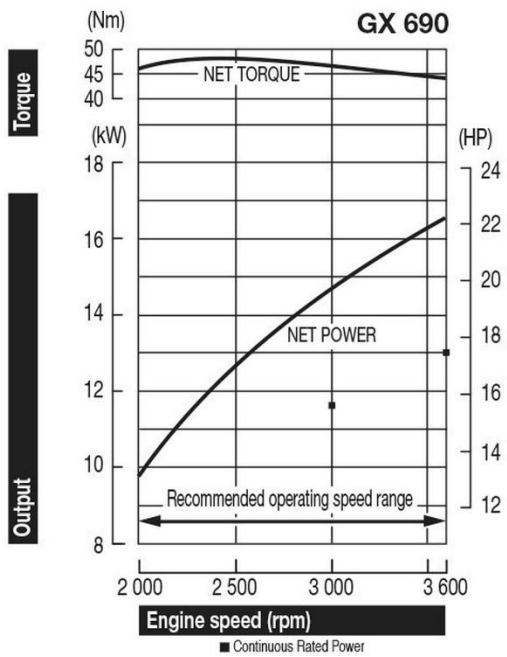

Figure 8. Manufactured specified engine torque and power vs speed for GX690 engine running with gasoline as fuel.

Methane Power Curves

Due to the late success of our methane conversion, extensive engine performance data was not collected. On our first run, a short power curve was created, with four torque values. Again, the performance of the engine was much worse than anticipated, likely due to inaccurate AFR. Once we reached a torque of 18 ft-lbs, the engine died and stopped the test.

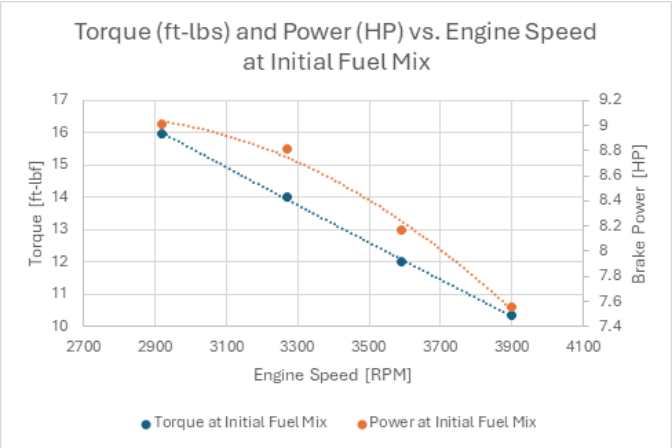

Figure 9. Engine torque and power vs speed for GX690 engine running with initial methane fuel mix.

After this first test, a similar, yet expedited method of tuning the AFR was performed. Another performance sweep was conducted, resulting in a ~16% power increase from 9-10.4 HP at peak power (2280 RPM).

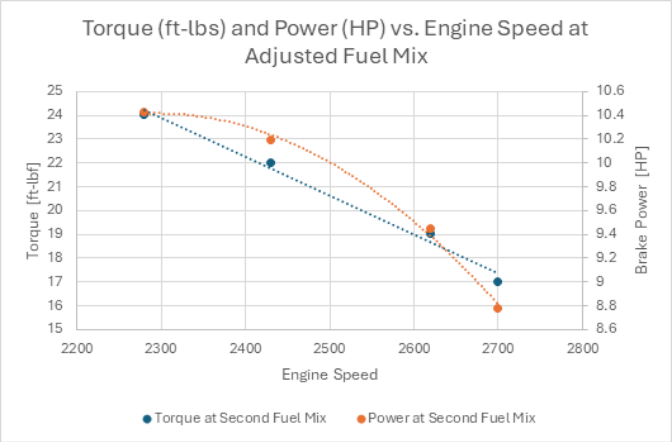

Figure 9. Engine torque and power vs speed for GX690 engine running with adjusted methane fuel mix.

As seen in the provided power curve from Honda for the GX690, the power is expected to increase as the engine speed increases in the range of 2000 to 3600 RPM. We did not see this trend. However, even if the propane mixture was running at a perfect stoichiometric mix, the same trend would be expected. This is because both fuels that are considered (propane and CNG) are gaseous fuels, while gasoline is a liquid. Because of the different state the fuel is in, there are expected to be greater losses, primarily in volumetric efficiency. This is because the fuels take up considerably more volume in the intake manifold and the cylinder than gasoline. Furthermore, at higher engine speeds there is a greater chance of misfires and incomplete combustion of the fuel, especially if the AFR is not as close to stoichiometric as possible. Due to these two considerations the power drops off at higher engine speeds is something to be expected when converting small, spark-ignited engines to operate on gaseous fuels.

Discussion

Despite the physical and thermochemical differences between propane and methane, we witnessed nearly identical performance between the two fuels in practice. Additionally, our values for output power were significantly lower than published values (from Honda) for the GX690 running on natural gas.

Table 4. Comparison of Manufacturer Performance Data to Our Own (Natural Gas)

Published Power value from Honda [bhp]	Value found in our own testing [bhp]
18.1	10.4

This means that we were almost certainly operating much further from stoichiometric air-fuel ratios than we had intended to. In either case, our only control of the air-fuel mixture was limiting the fuel supply pressure with various pressure control valves. Manual adjustment of supply pressure led to large, imprecise changes in AFR. In one case with methane a small quarter turn of the fuel adjustment screw resulted in the engine shutting off instantly.

Other considerations would be adjusting the timing of the engine to accommodate the fuel switch. Due to the higher octane and lower flame speeds of both propane and CNG, more advanced timing would allow for increased performance.

Additionally, our only method for estimating our equivalence ratio for methane was qualitative auditory feedback from the engine, which is obviously far from ideal. If the exhaust gas analyzer was operational at the time of testing, we could have input our fuel on the device and it would have displayed our equivalence ratio, λ, in real time on its display.

Based on our measured performance data, it is reasonable to assume that we were operating at equivalence ratios far from unity in the case of both propane and methane.

This is what led to us seeing similar output performance from both fuels, and in both cases being significantly below the rated power numbers from Honda.

Summary/Conclusions

In this project we successfully converted a Honda GX690 to run on both propane and CNG as an alternative to gasoline. Furthermore, we were able to successfully mount the engine on a dynamometer in the lab and get data. Initial data suggests that both fuel conversions result in a lower power output than gasoline but are more economical. Robust storage and delivery infrastructure is required due to the nature of pressurized gaseous fuels, which is a challenge for certain applications such as motor vehicles. However, there are several factors that were not investigated in this lab and could be further investigated to determine the validity of pursuing fuel conversions on small spark ignition engines. One issue that really limited us was that the exhaust gas analyzer device was not operational on the days of our testing. Without this we were unable to tune our fuel flow to a stoichiometric mixture, which explains some of the performance variations that were observed. Additionally, we were unable to pursue the question of whether these alternative fuels would change emissions from an environmental standpoint.

References

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2. Honda Motor Co. *GX690 Owner's Manual*. Tokyo: Honda Motor Co., Ltd., 2010. Accessed June 1, 2025. Honda GX690 Manual.
3. Sahoo, B.B, et al. "Effect of Engine Parameters and Type of Gaseous Fuel on the Performance of Dual-Fuel Gas Diesel Engines-A Critical Review." Renewable and Sustainable Energy Reviews, Pergamon, 1 Oct. 2008, www.sciencedirect.com/science/article/pii/S1364032108001020 .

Definitions/Abbreviations

CNG	compressed natural gas
GGE	gasoline gallon equivalent
AFR	air fuel ratio
HP	horsepower
BHP	brake horsepower

Appendix

Conversion Disassembly Instructions

Conversion Steps Back to Gasoline:

1. Remove plastic airbox cover.
2. Remove wingnut and air filter.
3. Remove the outer four bolts.
4. Remove bottom half of the airbox cover.
5. Detach carburetor and gasket.
6. Reassemble without bolt spacers.

Contact Information

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